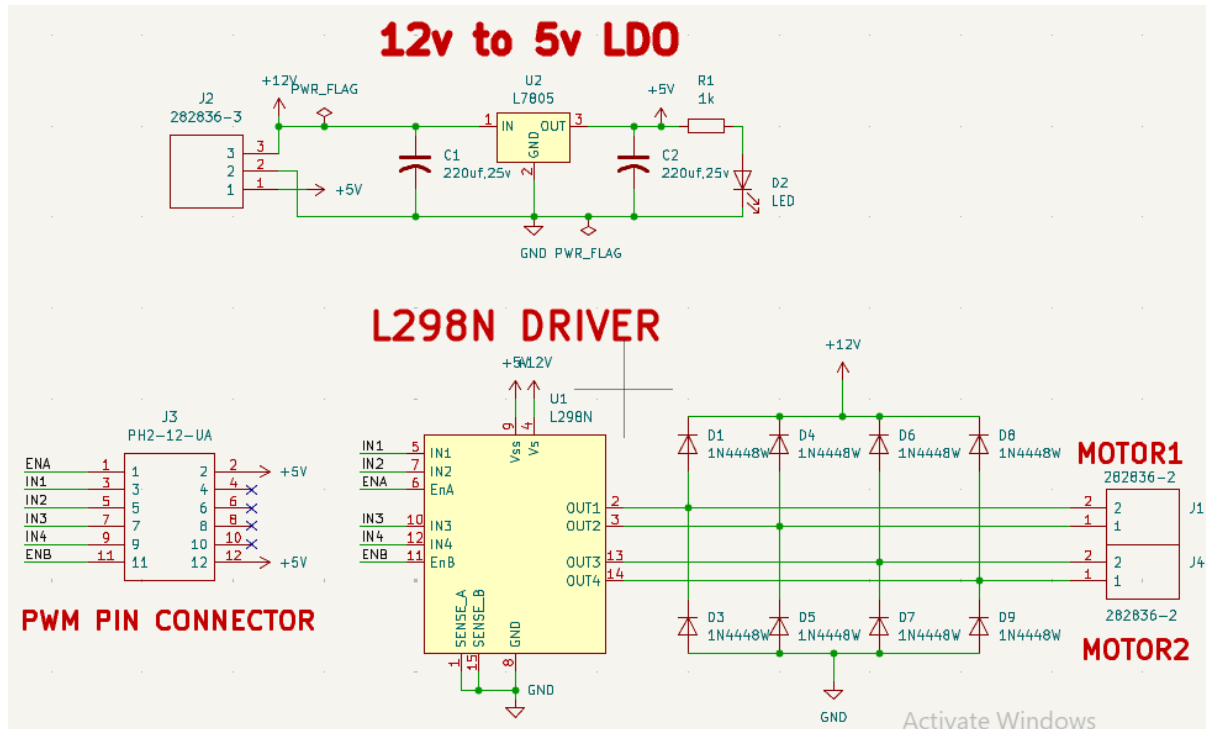


Dc Motor L298n H-Bridge

A practical, step-by-step user manual that walks you from a blank project to production-ready Gerbers using KiCad. Intended for beginners and intermediate users who want a clean, repeatable PCB design workflow.

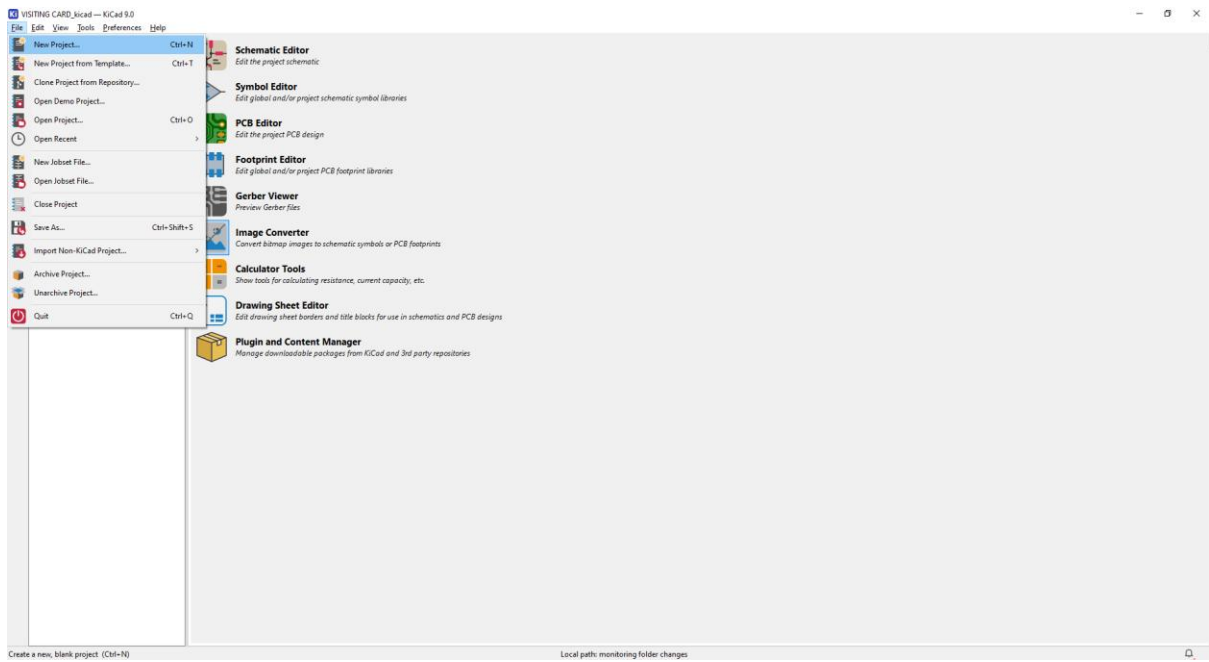
Circuit diagram:



Start a project:

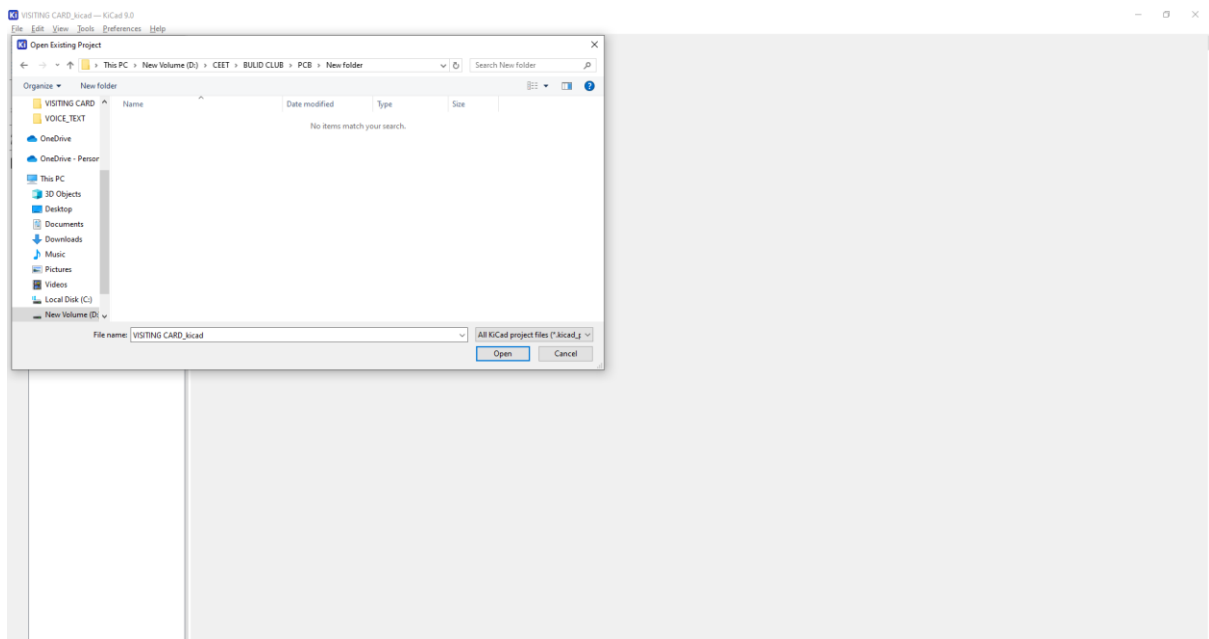
Step1:

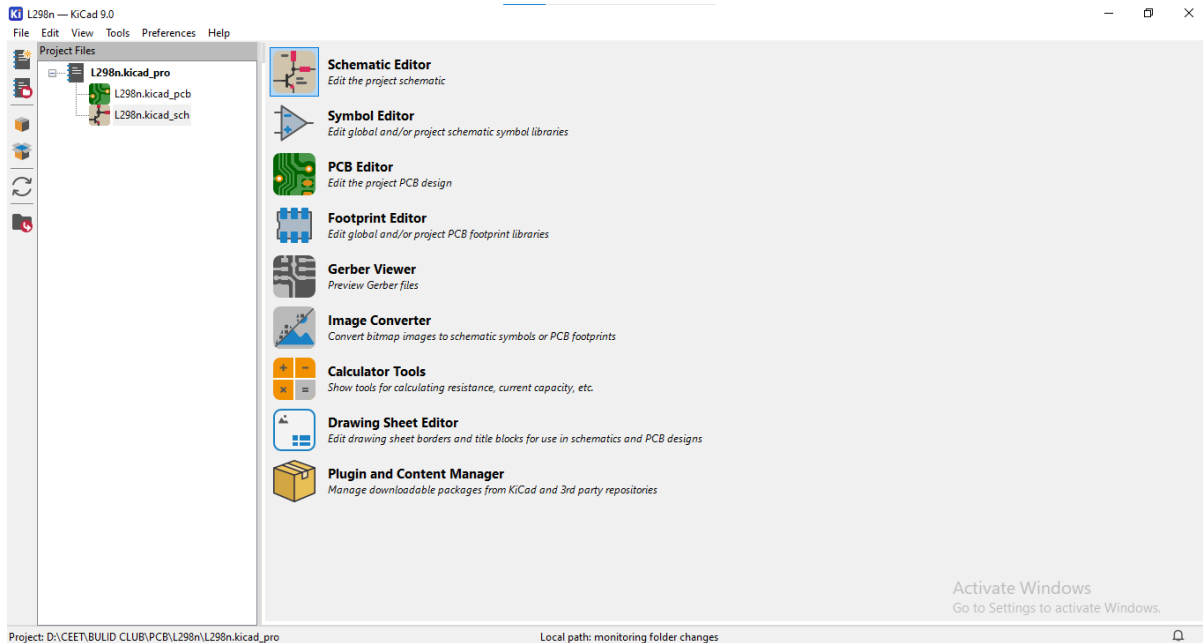
- File → New Project → give a clear name



Step2:

- Enter the project title
- Save in a folder structure.





Step 3:

- Circuit components:

S. No	Components	Quantity
1.	L298n	1
2.	1n4448w Diode	8
3.	1k 0603 Resistor	1
4.	Led 0603	1
5.	Capacitor 220uf ,25v	2
6.	7805 voltage regulator	1
7.	PH2-12-UA Berg strip male connector	1
8.	282836-2 motor connector	2
9.	282836-3 motor connector	1

Some components are not there in the kicad Ide so externally download the components after import the kicad ide.

- L298n H-Driver IC download- [link](#)
- 7805 voltage regulator download- [link](#)
- PH2-12-UA Berg strip male connector download- [link](#)
- 282836-2 motor connector download- [link](#)
- 282836-3 motor connector download- [link](#)

Go to download the All components.

- Create a mouser Account -[link](#)

The screenshot shows the 'Account Registration' page on the Mouser website. At the top, there is a navigation bar with 'EXAMPLES', 'LOGIN', and 'REGISTER' links. The main heading is 'Account Registration'. Below it, a paragraph states: 'Supplyframe/SamacSys is a third-party service provider to Mouser and a data controller that is collecting customer personal data for the purposes of updating you about component updates and changes to CAD models. Please follow the link to the Supplyframe/Samacsys [Privacy Policy](#) for additional information.'

The registration form includes the following fields:

- Title: ** Pick One ** (dropdown)
- First Name: text input
- Last Name: text input
- Company: text input
- Country: ** Pick One ** (dropdown)
- State: Choose Country First (dropdown)
- Town: Find Me (button) and Choose State/County First (dropdown)
- Job Role: ** Pick One ** (dropdown)
- Email: text input
- Password: text input
- Confirm Password: text input
- Public Alias: text input

Below the form, there is a checkbox for 'I agree to the [Terms](#) and [Privacy Policy](#)' and a 'Register' button. On the right side of the page, there is a 'Activate Windows' watermark with the text 'Go to Settings to activate Windows.'

- Account Registration ones completed after going to login it

The screenshot shows the Mouser website's login page. The browser address bar displays 'ms.componentsearchengine.com/ga'. A 'Sign in' modal is open in the center of the screen, with the URL 'https://ms.componentsearchengine.com' at the top. The modal contains two input fields: 'Username' and 'Password'. Below the fields are two buttons: 'Sign In' (in blue) and 'Cancel' (in light blue). The background of the page is dark gray. At the bottom right, there is a 'Activate Windows' watermark with the text 'Go to Settings to activate Windows.'

- Example: IRLZ24NPBF Mosfet download- [link](#)



Images are for reference only
See Product Specifications

[Share](#)

Mouser No:
942-IRLZ24NPBF

Mfr. No:
IRLZ24NPBF

Mfr.:
[Infineon Technologies](#)

Customer No:

Description:
MOSFETs MOSFT 55V 18A 60mOhm 10nC LogLvIAB

Lifecycle:
 Obsolete

Datasheet:
[IRLZ24NPBF Datasheet \(PDF\)](#)

ECAD Model:

 PCB Symbol, Footprint & 3D Model
 Download the free [Library Loader](#) to convert this file for

Stock:
[Restricted Availability](#)

Possible Replacement



Mfr.: Infineon Technologies
Mfr. No: [IRFB7545PBF](#)
Description: MOSFETs MOSFET N CH 60V 95A TO-220AB

[Compare With Current Part](#)

FEATURED PRODUCTS
INFINEON



OptiMOS™ 6 80V Power MOSFETs
 Sets industry benchmark performance with a wide portfolio offering.
[Learn More](#)

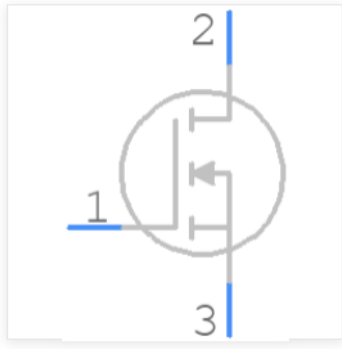
- Click the ECAD Model option

Symbols | Footprints | 3D Models

Infineon IRLZ24NPBF

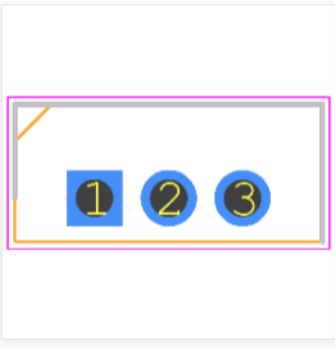
MOUSER ELECTRONICS

 **Symbol**

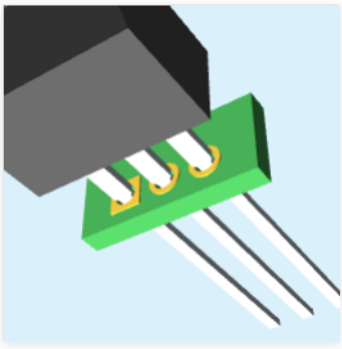


Normal View

 **Footprint**



 **3D model**



Device on Footprint

[Download CAD Models](#) 

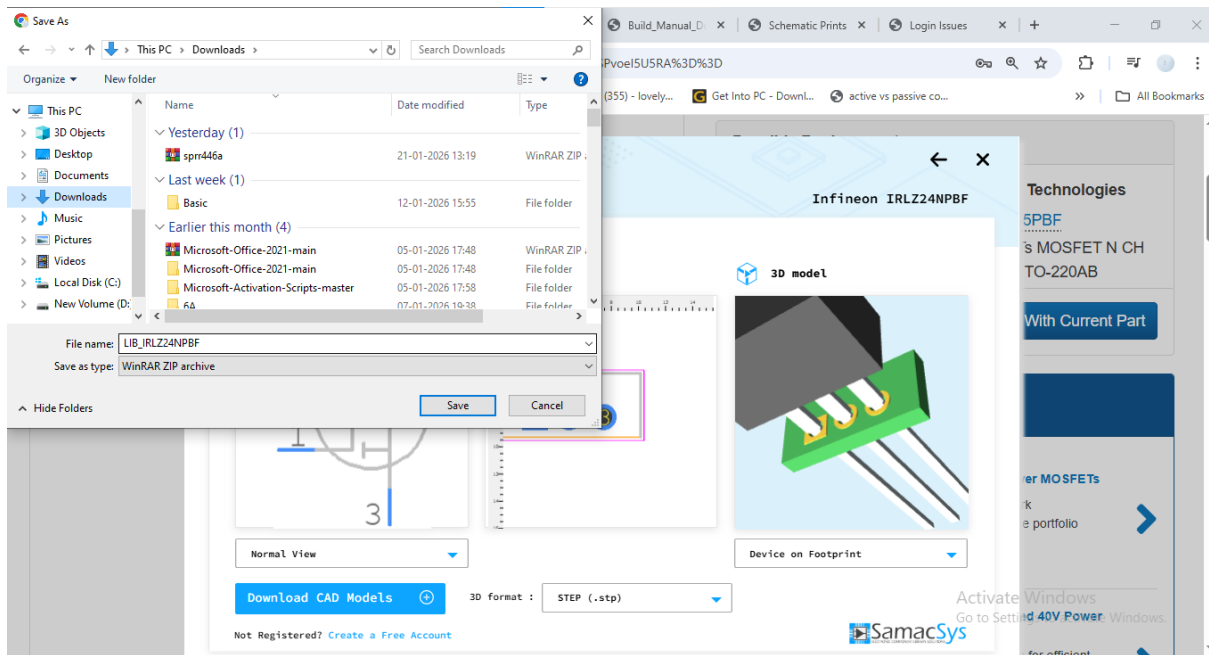
3D format :

Not Registered? [Create a Free Account](#)

Activate
Go to Setti

SamacSys
ELECTRONIC COMPONENT LIBRARY SOLUTIONS

- Click the Download CAD Model option



- Save the Zip file.

Step 4:

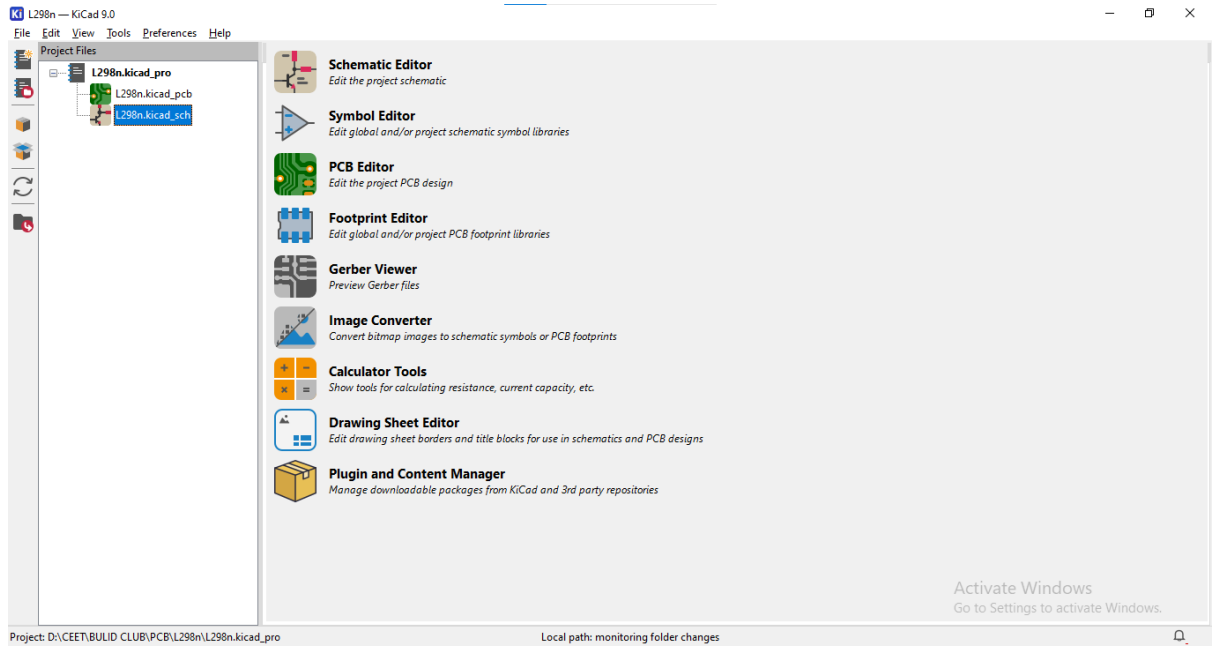
- Download the components after extracting the ZIP file.
- Go to Preferences > Manage **Symbol Libraries** Open it
- Add existing library to table option has been open it.
- Open the components folder, select the kicad folder
- Select the Kicad Library file
- After Symbol Libraries window has ok option click it.

Step 5:

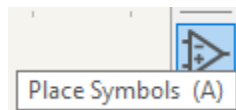
- Go to Preferences > Manage **Footprint Libraries** Open it
- Add existing library to table option has been open it.
- Select the legacy
- Open the components folder, select the kicad folder.
- Select the Kicad Library file
- After Footprint Libraries window has ok option click it.

Step 6:

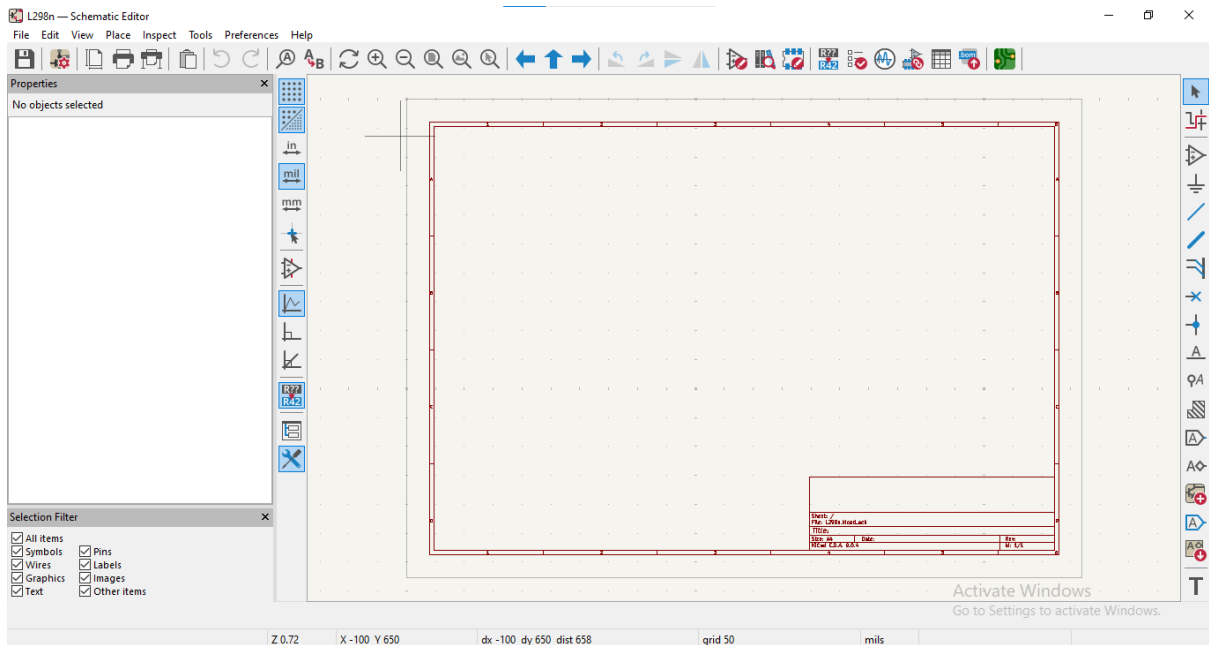
- Open the schematic editor file



- Click the Place Symbols ion,



- On your right-side top corner / press 'A' key word

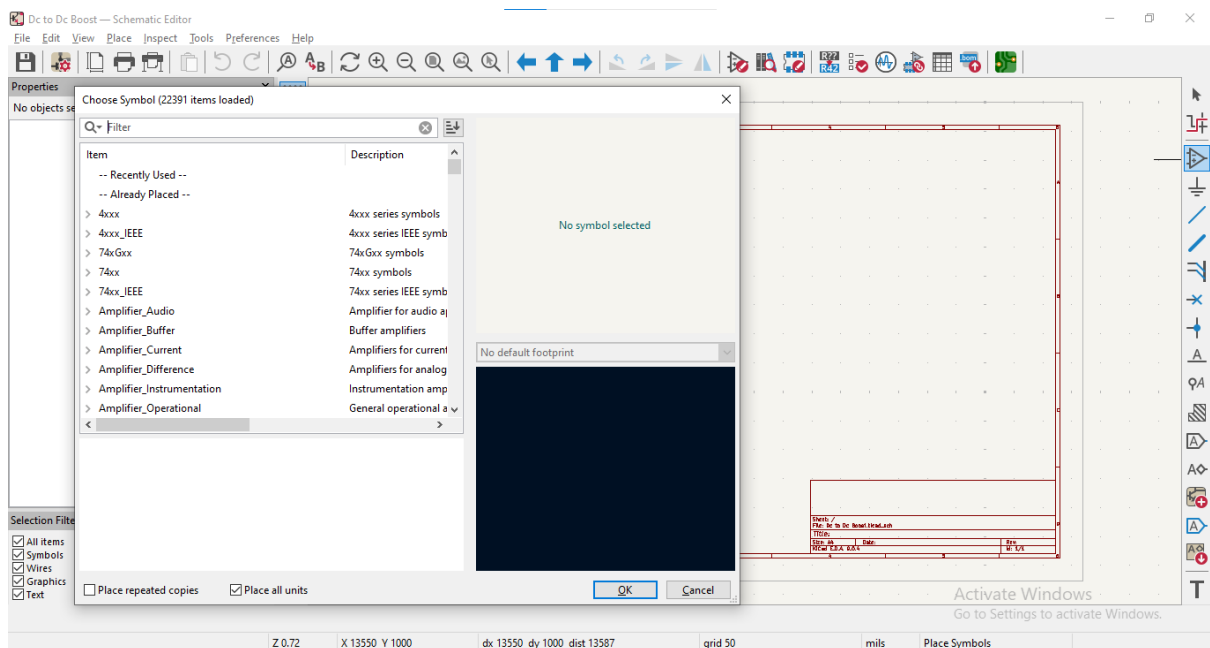


Step 7:

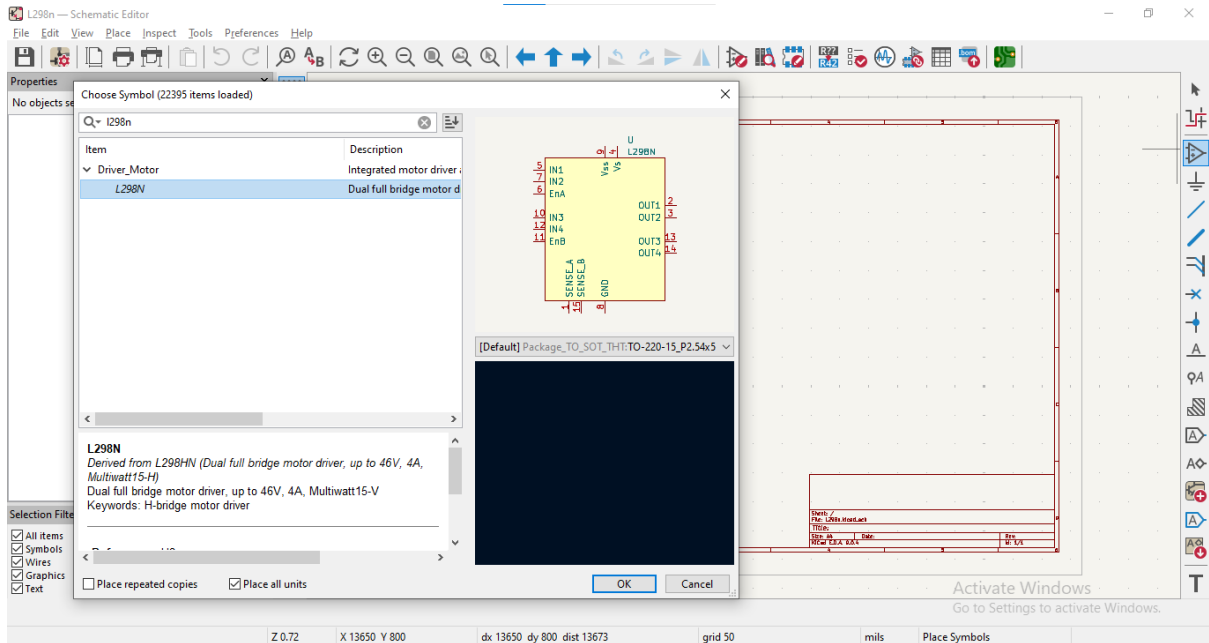
- Select your circuit components

S. No	Components	Quantity
1.	L298n	1
2.	1n4448w Diode	8
3.	1k 0603 Resistor	1
4.	Led 0603	1
5.	Capacitor 220uf ,25v	2
6.	7805 voltage regulator	1
7.	PH2-12-UA Berg strip male connector	1
8.	282836-2 motor connector	2
9.	282836-3 motor connector	1

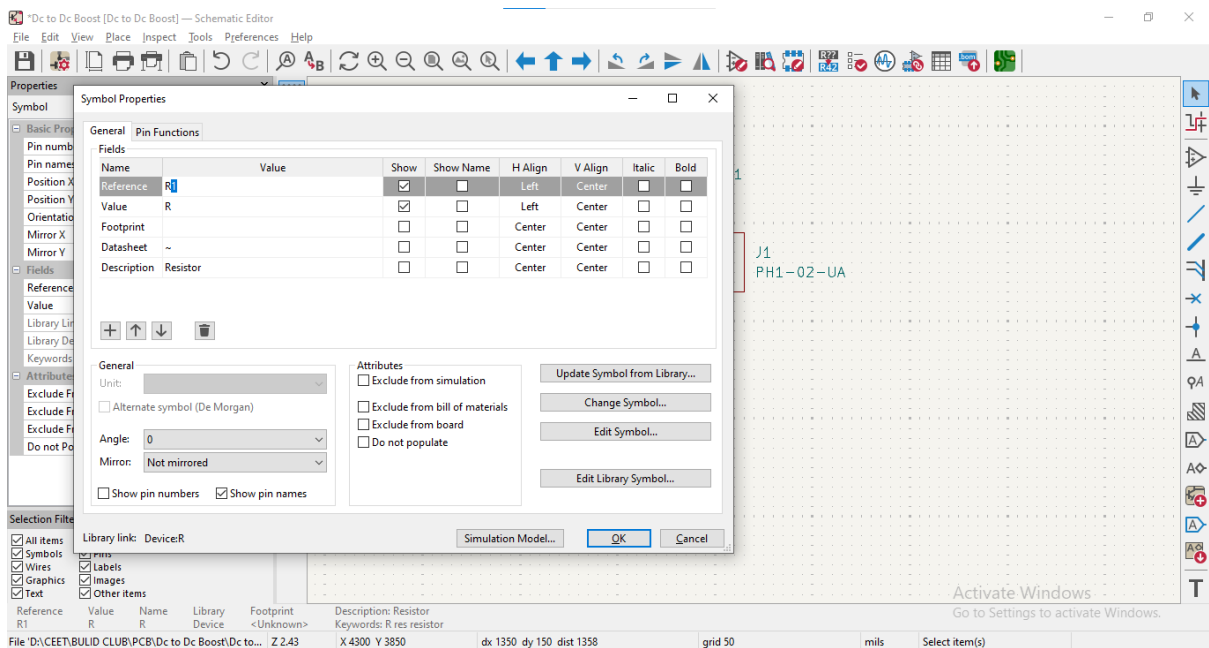
- Go to filter option
- Search the components name or part number (ex-Resistor or L298n)



- Select your component after placing it



- To place all components on the sheet
- Select the component press 'E' add the value (Resistor, inductor) and all components add the footprint



- Give the value like that (2.2k ohms or 100uf)

*Symbol Properties

General Pin Functions

Fields

Name	Value	Show	Show Name	H Align	V Align	Italic	Bold
Reference	R1	☒	☐	Left	Center	☐	☐
Value	2.2K	☒	☐	Left	Center	☐	☐
Footprint		☐	☐	Center	Center	☐	☐
Datasheet	~	☐	☐	Center	Center	☐	☐
Description	Resistor	☐	☐	Center	Center	☐	☐

+ ↑ ↓ 

General

Unit:

☐ Alternate symbol (De Morgan)

Angle:

Mirror:

☐ Show pin numbers ☒ Show pin names

Attributes

☐ Exclude from simulation

☐ Exclude from bill of materials

☐ Exclude from board

☐ Do not populate

Update Symbol from Library...

Change Symbol...

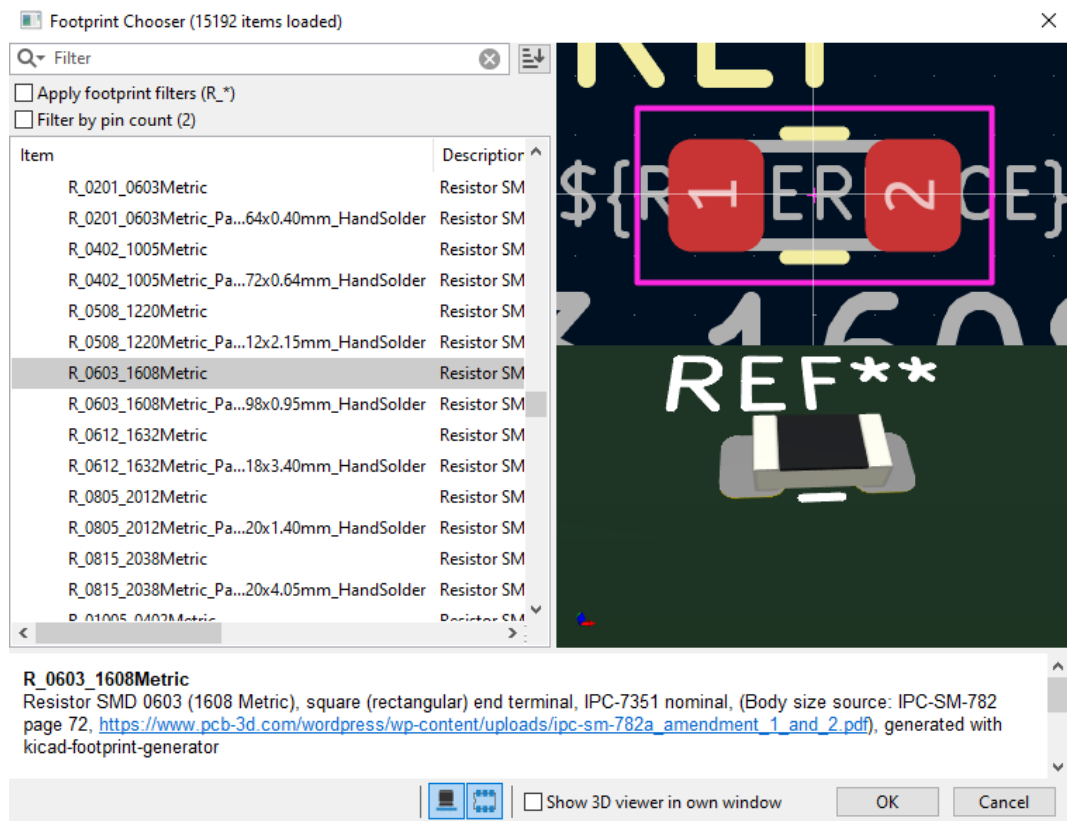
Edit Symbol...

Edit Library Symbol...

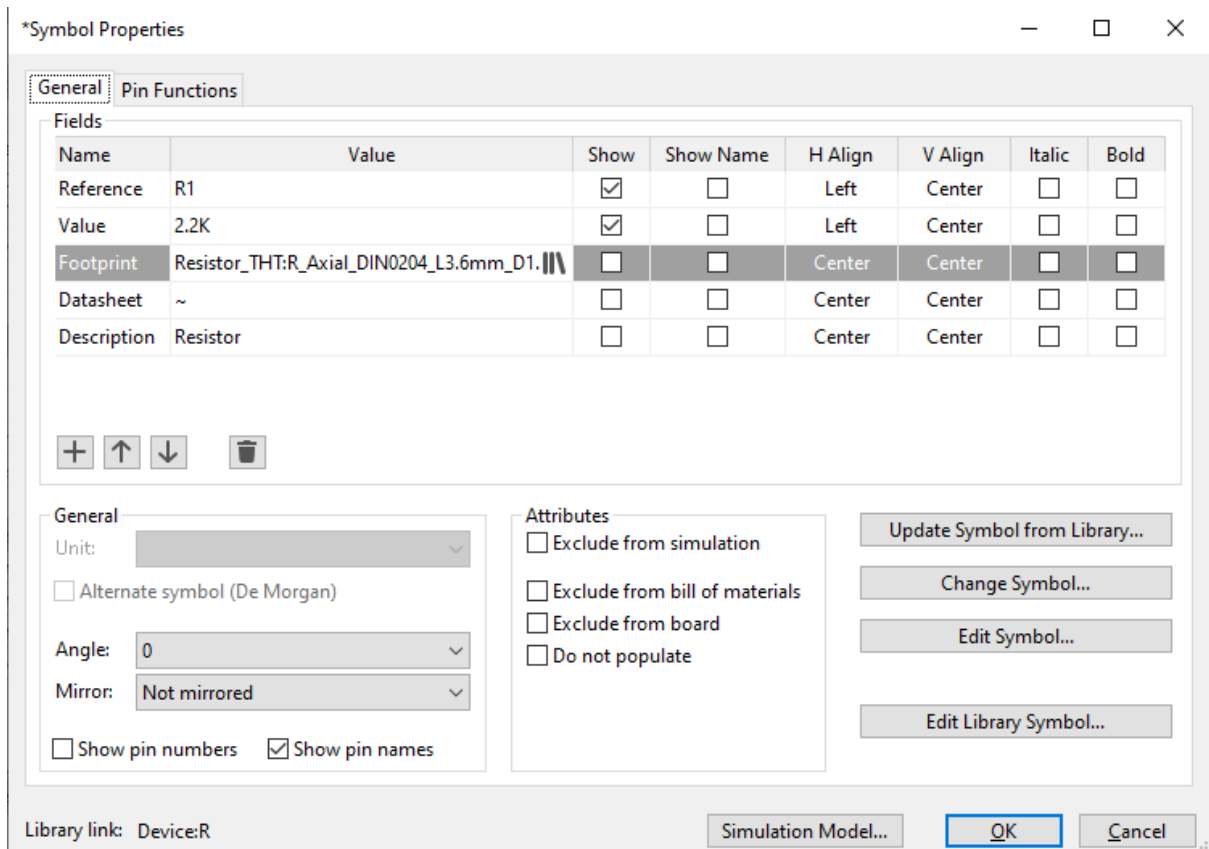
Library link: Device:R

Simulation Model... OK Cancel

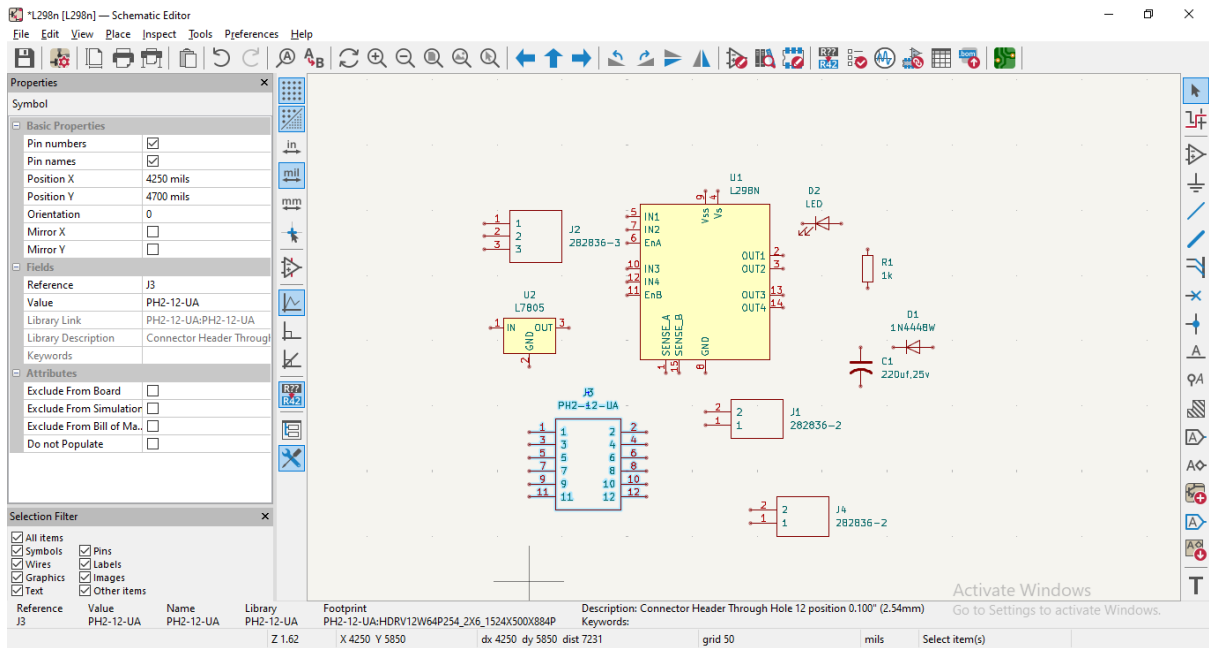
- Press the footprint option
- Select the part number (like a package-through hole)
- Go to filter option
- Search the components name or part number



- Select your component Footprint after ok it
- Symbol Properties finally ok option clicks



- Next give all components value has enter symbol properties

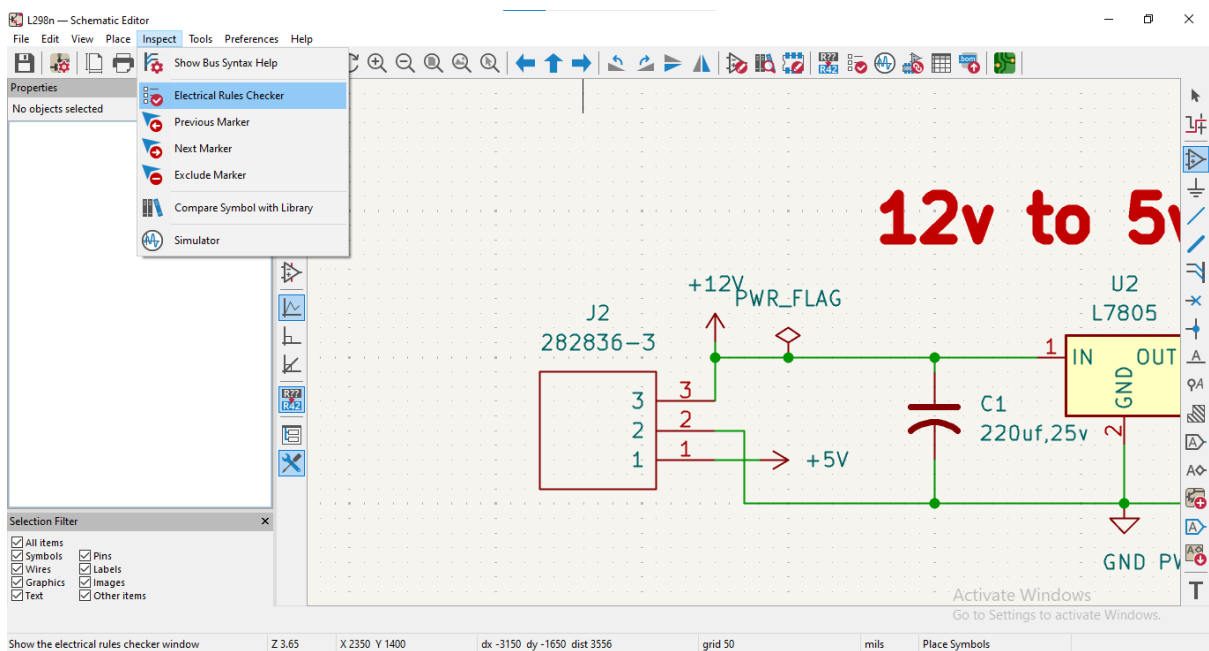


- Additionally power & gnd pin to connector the PWR_PLAG Connect symbols with wires
- Press W key word, go to connect the wires
- Save the Schematic file.

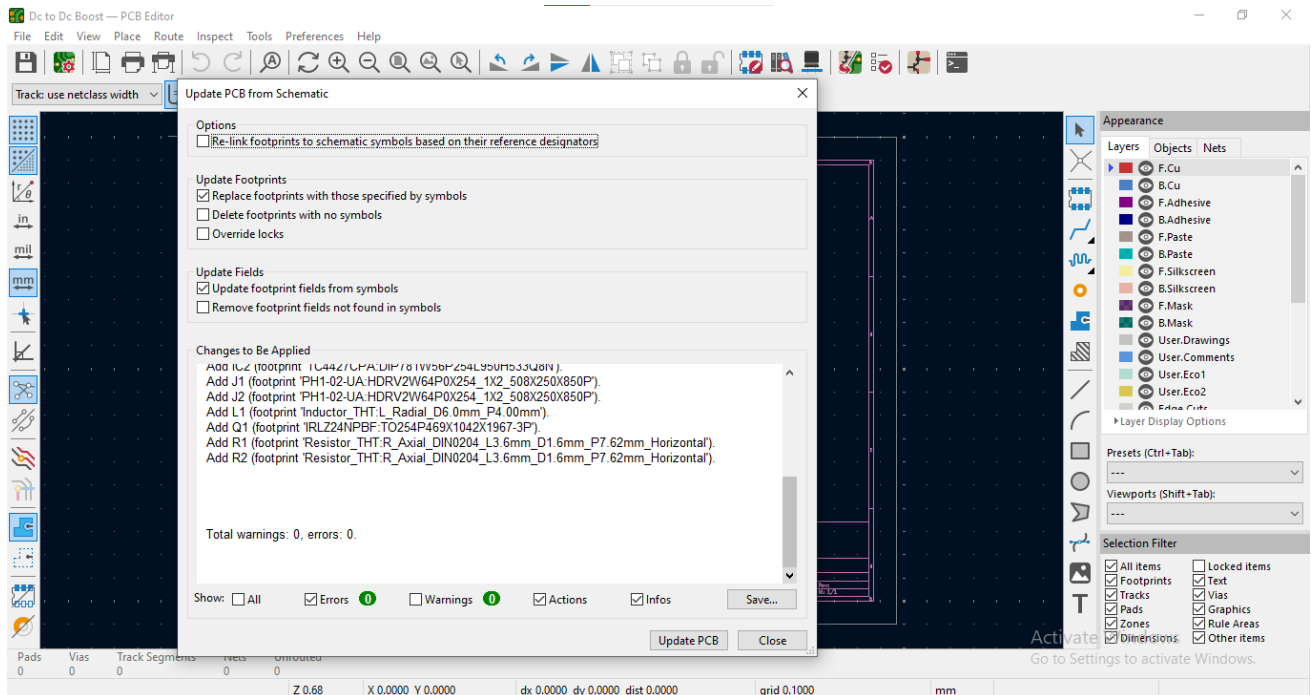


Step 8:

- Go to 'Tool' → 'Update PCB from schematic' / F8



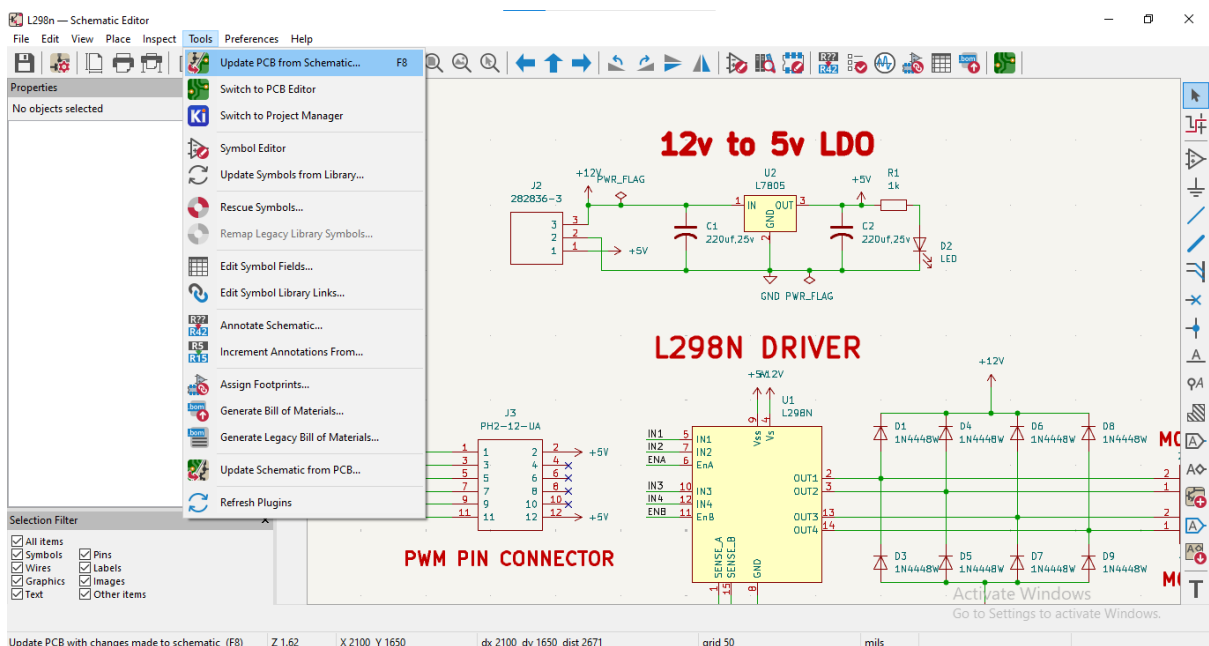
- Check each schematic symbol must have a matching PCB footprint
- If any footprint has missing it can't update PCB, it's shown error warning



- Go to clear the error (like a No unconnected, annotate symbols, fix wiring errors, missing power flags) Otherwise update the PCB option click

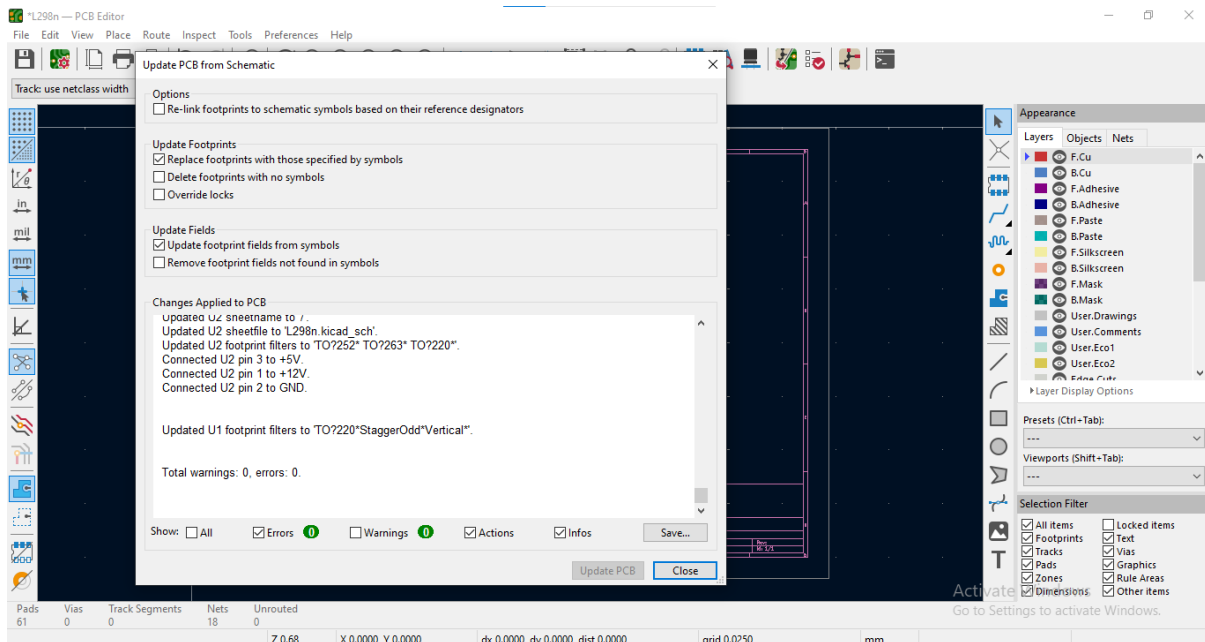
Step 8:

- Go to 'Tool' → 'Update PCB from schematic' / F8



- Check each schematic symbol must have a matching PCB footprint

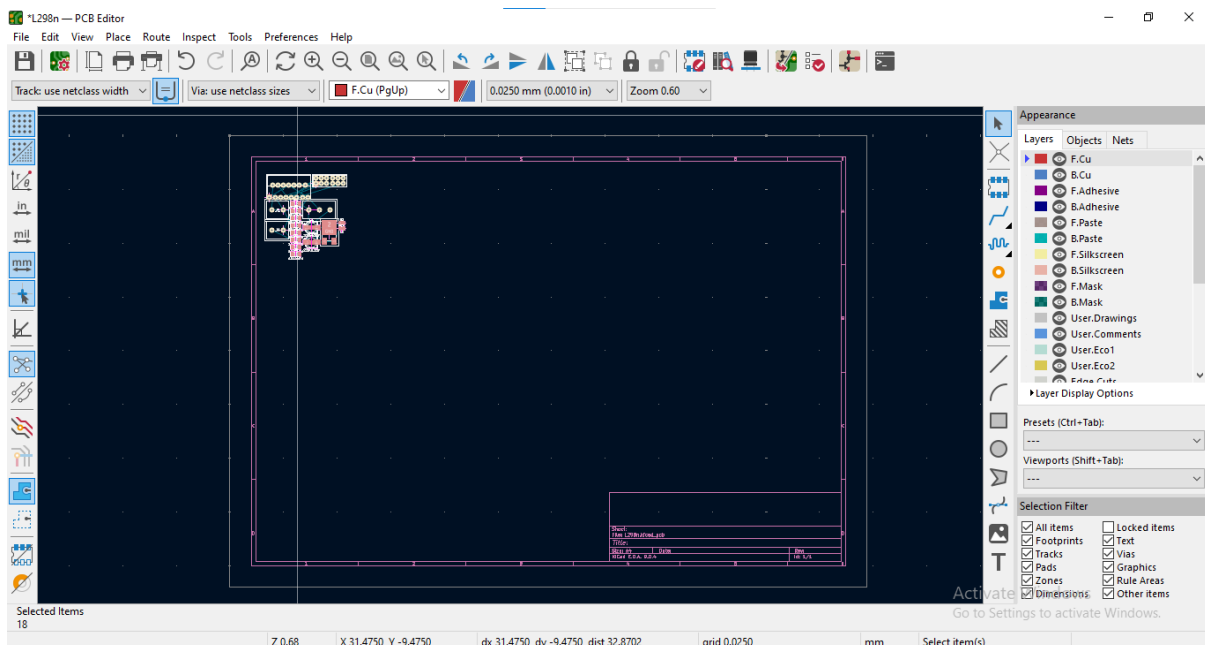
- If any footprint has missing it can't update PCB, it's shown error warning



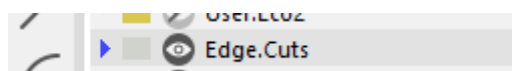
- Go to clear the error (like a No unconnected, annotate symbols, Fix wiring errors, missing power flags) Otherwise update the PCB option click

Step 9:

- Set the board outline (Edge Cuts layer)

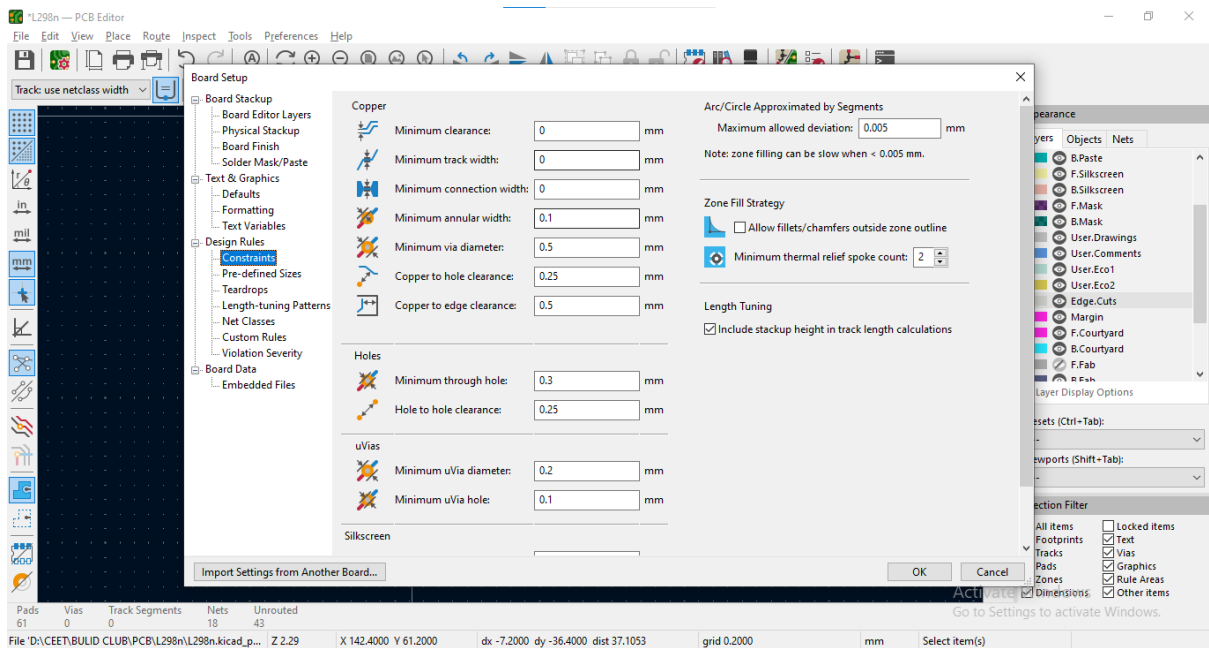
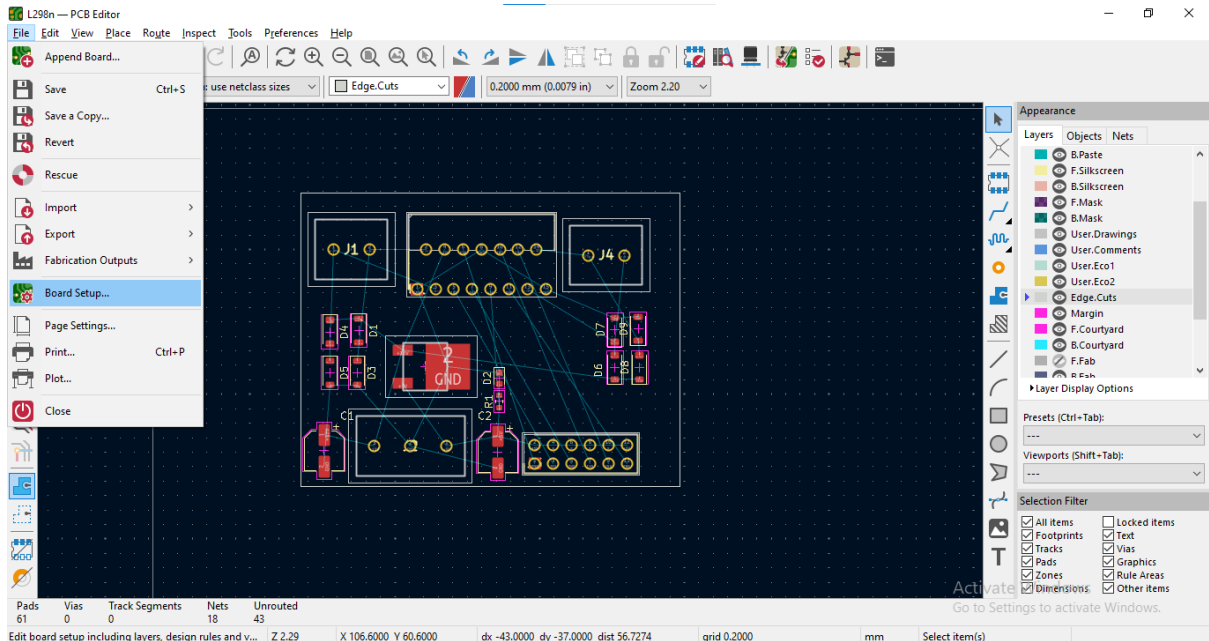


- Click the Place Symbols icon, On your right-side corner



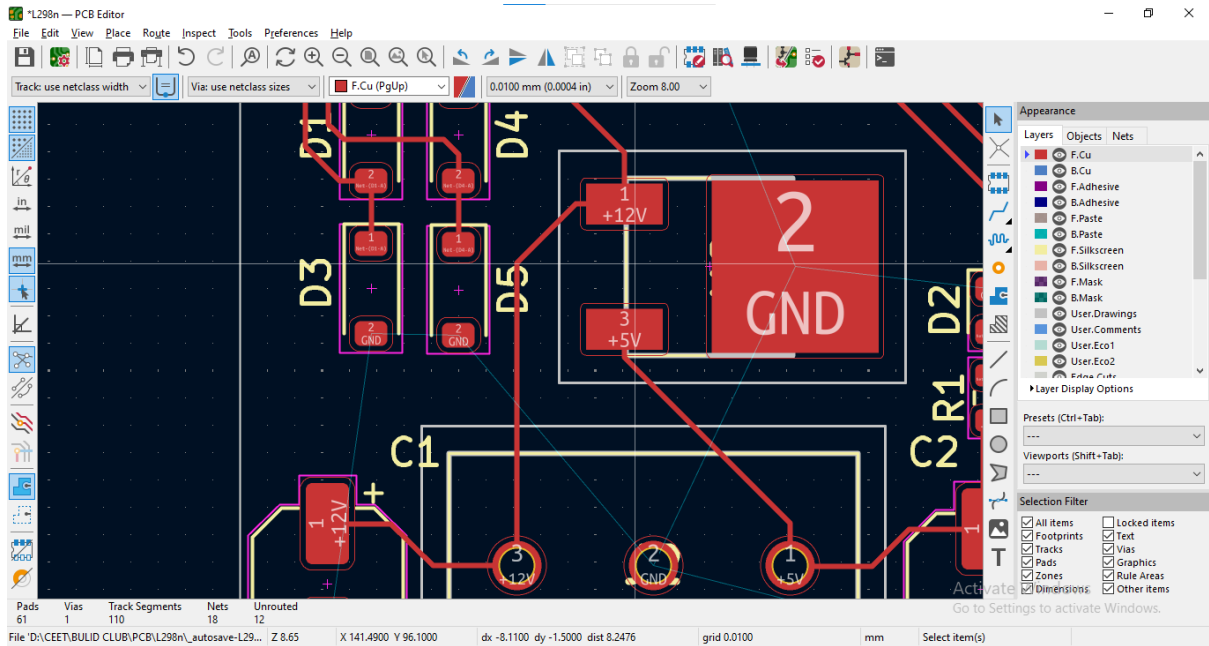
-
- The screenshot shows the KiCad PCB Editor interface. The top menu bar includes File, Edit, View, Place, Route, Inspect, Tools, Preferences, and Help. The toolbar contains icons for file operations, navigation, and editing. The main workspace shows a dark blue grid with a white rectangular selection box. The bottom status bar displays coordinates and dimensions. The right sidebar shows the 'Appearance' panel with a list of layers and display options.

- Next go to File → Board Setup → Constraints
- Set the details of all option
- And refer the [link](#) (**Disclaimer** – ‘Constraints’ depends on PCB manufacturer)

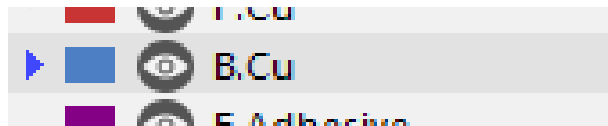


Step 11:

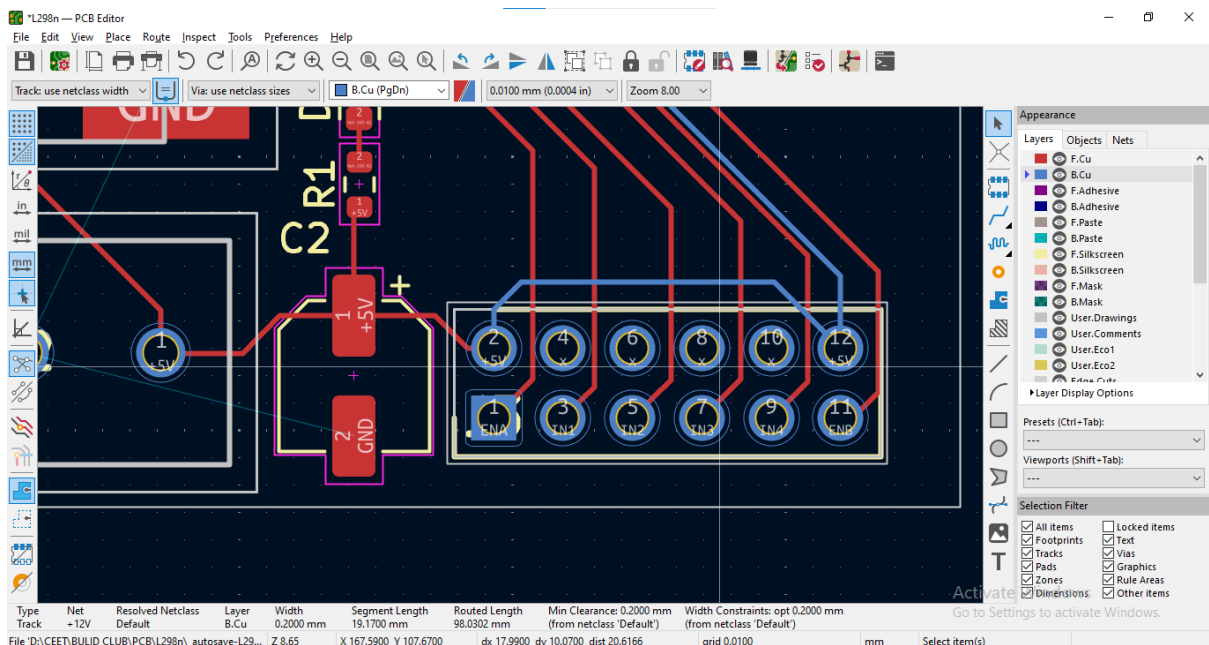
- Go to select the layer after routing in the traces short key “X”
- Traces thicker increase or decrease width via net class selection based



- Go to select the bottom layer
- Click the Place Symbols ion, On your right-side corner



- Go to select the layer after routing in the traces short key “X”



- If need VIA Click the Place Symbols ion, On your right-side corner

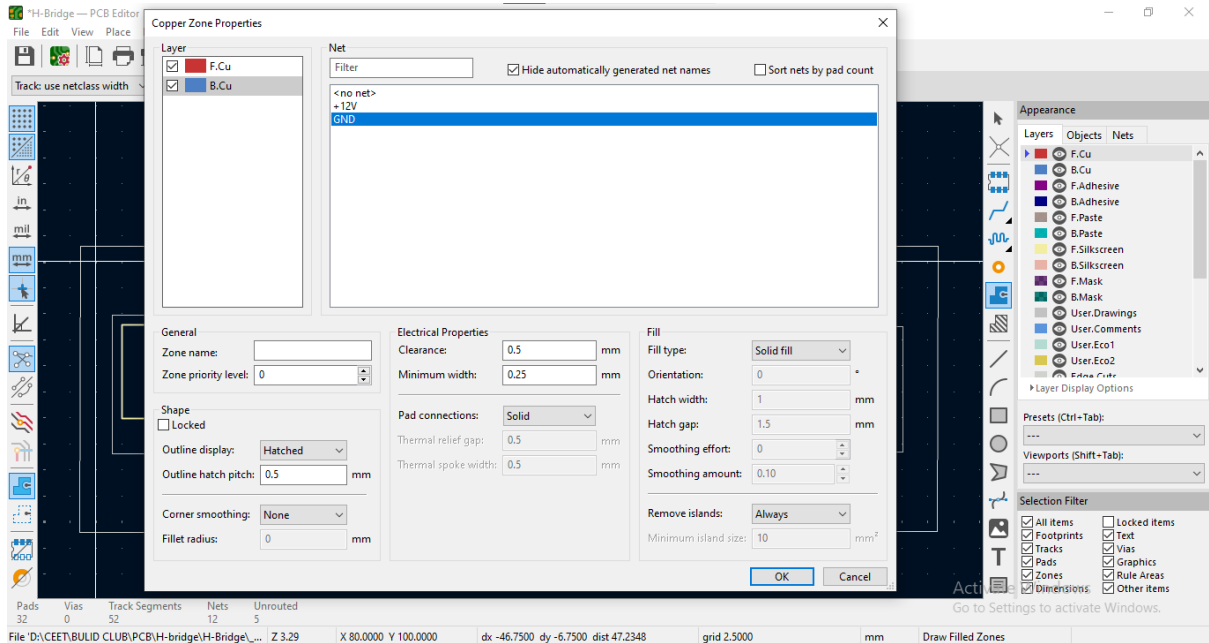


- Click the Place Symbols icon, after click the PCB grid

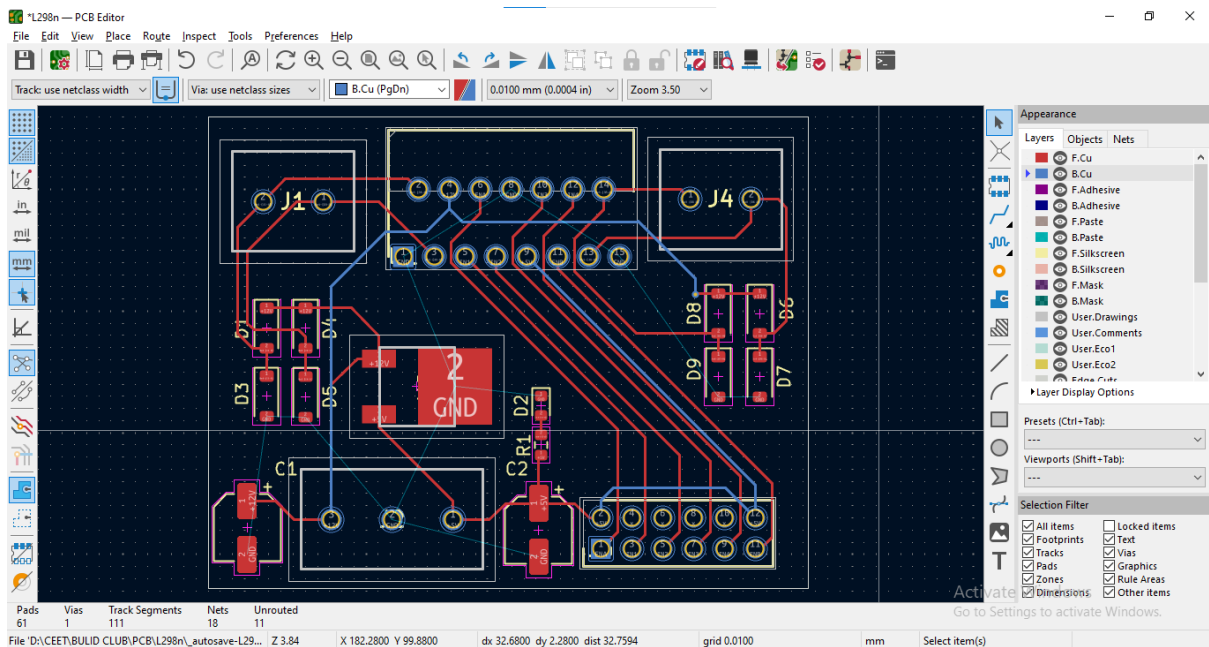


Polygon tool

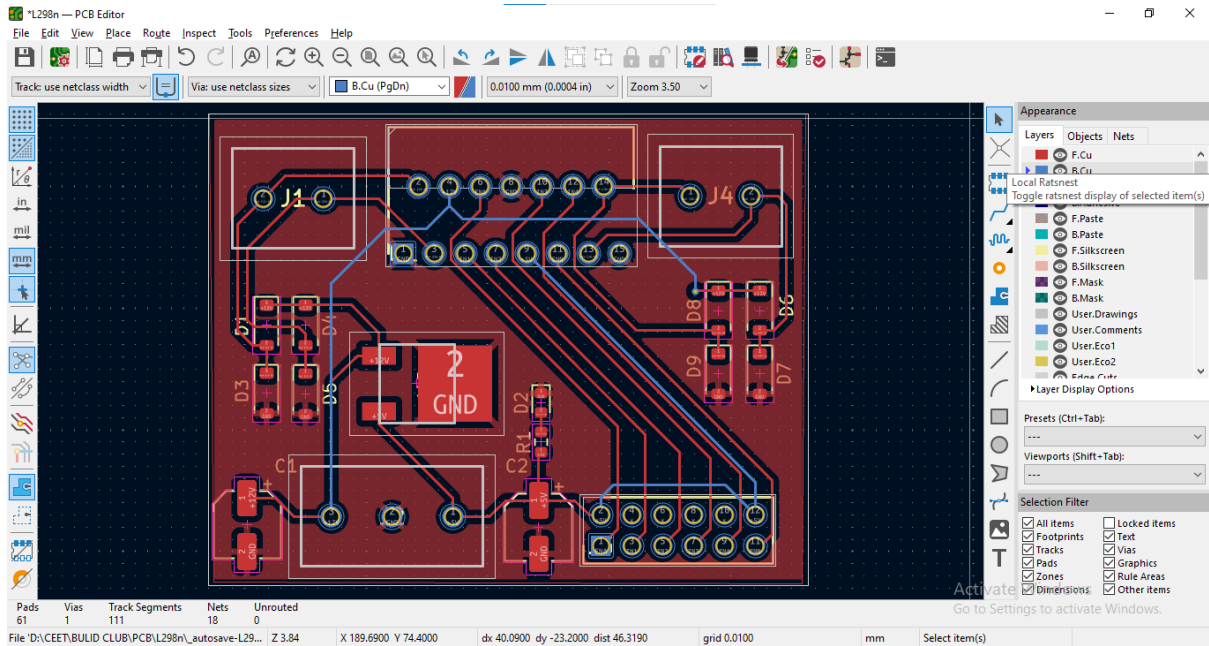
- It's automatically open Copper Zone Properties, selected the GND net



- Polygon using to draw the rectangle

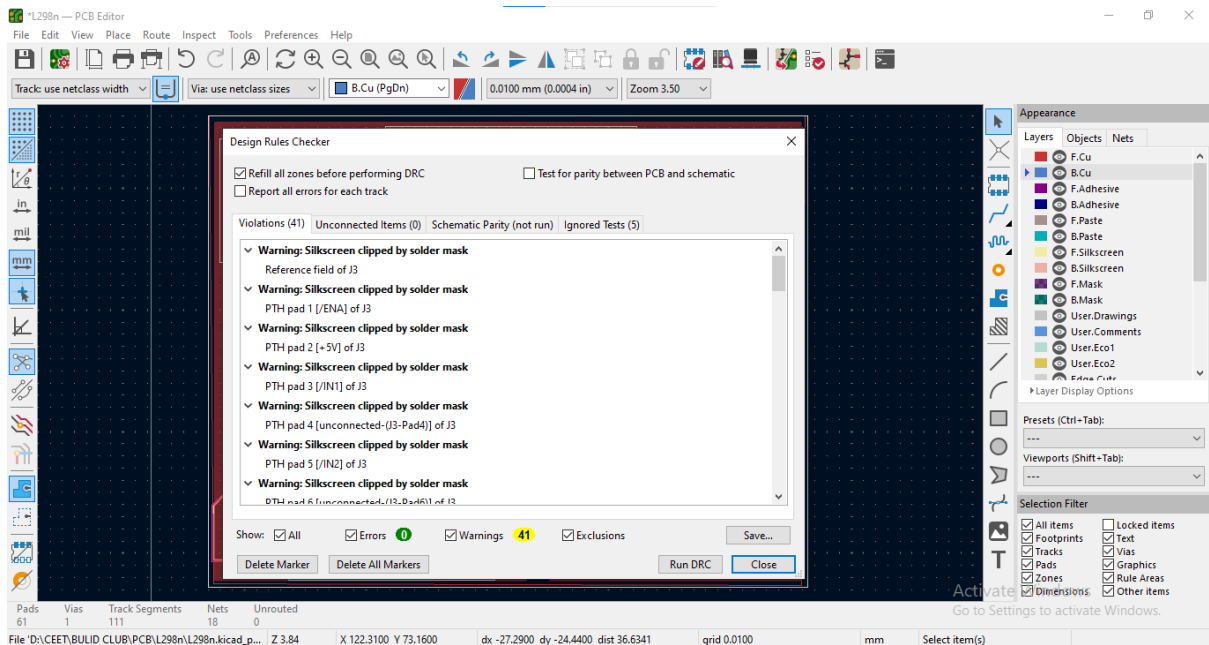


- Press the key “b” or go to edit option → Fill All Zones click it



Step 12:

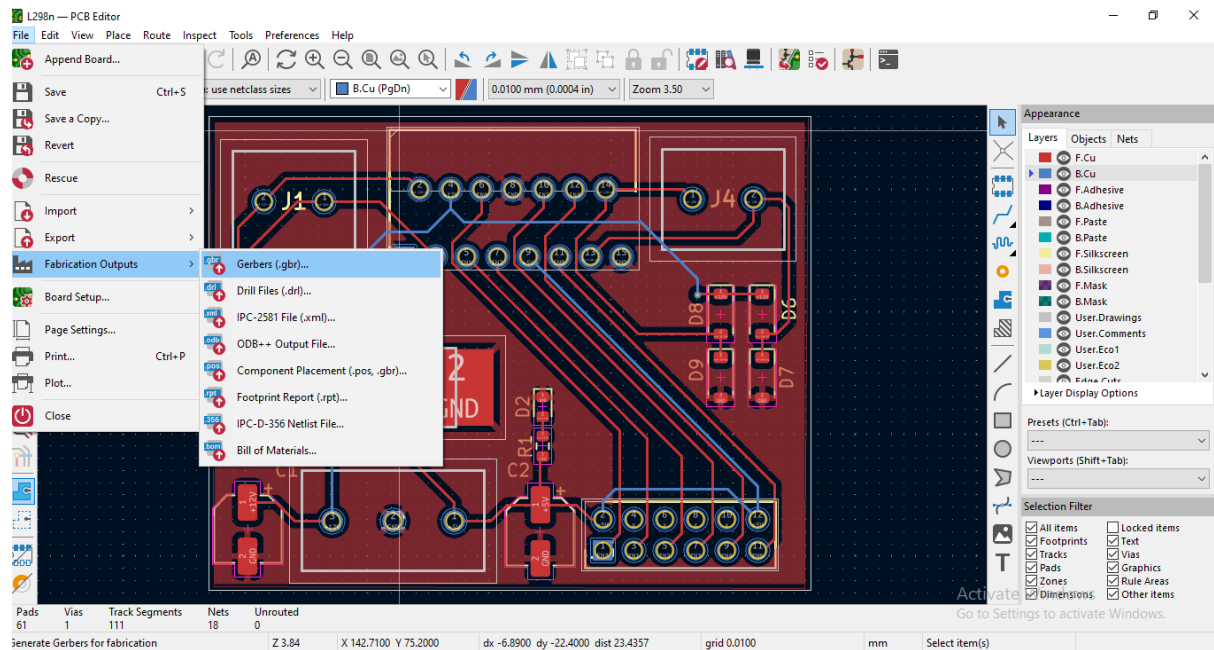
- Go to 'Inspect' → 'Design Rules checker'
- Check each Constraints value must have a matching PCB design
- If any Constraints value has missing it can't run DRC, it's shown error warning Go to clear the error (clearances, unconnected pins, minimum annular rings)



- If clear all error go to Fabrication outputs

Step 13:

- Go to file → Fabrication outputs → Gerber, click it



- Select all layers
 - F. Cu (Top copper)
 - B. Cu (Bottom copper)
 - F. Silk (Top silkscreen)
 - B. Silk (Bottom silkscreen)
 - F. Mask (Top solder mask)
 - B. Mask (Bottom mask)
 - Edge. Cuts (Board outline)
 - Drill file (Excellon)
- Next Click the Generate Drill Files option

